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Direct microwelding and sealing of glass-ceramics with burst-mode ultrafast laser

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Abstract

Glass-ceramics, a type of transparent hard and brittle material with ultra-low thermal expansion coefficients, find extensive applications in aerospace, precision optoelectronics, and other fields. Research on high-efficiency and high-quality welding and sealing technologies for glass-ceramics has become a key focus of current attention. In this work, ultrafast laser micro-welding technology was applied to the direct-write selective welding and sealing of glass-ceramics. Using burst-mode ultrafast lasers, we systematically investigated the effects of pulse duration within bursts, pulse energy, welding speed, and welding line patterns on interfacial material fusion. Furthermore, direct-write sealing of glass-ceramic optical chambers was performed using grid-

Outline

Abstract

Keywords

1. Introduction

2. Experiment

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4. Conclusion

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structured weld lines with burst-mode ultrafast lasers. With the surface flatness of samples being only one wavelength at 632 nm, a sealing leak rate of 10^{-7} Pa·m³/s was achieved. These results demonstrate that this technology exhibits significant advantages in the field of high-efficiency, high-quality, and chemical contamination-free welding and sealing of optical chambers.

Keywords

Ultrafast laser processing; Laser welding; Glass-ceramics

[Declaration of competing interest](#)

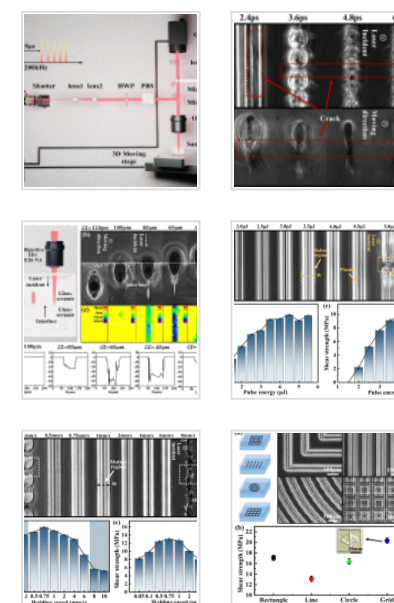
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1. Introduction

Glass-ceramics, a novel composite material prepared via glass crystallization processes, comprise both glassy and microcrystalline phases. This unique structure synergistically combines the optical transparency of glasses with the mechanical robustness of polycrystalline ceramics, while offering exceptional thermal stability (near-zero coefficients of thermal expansion in certain compositions), chemical corrosion resistance, and tunable physicochemical properties via tailored crystalline phase engineering [[1], [2], [3]]. Compared to conventional glasses, glass-ceramics exhibit significantly enhanced mechanical strength, superior thermal shock resistance, and outstanding wear resistance, rendering them highly resilient to mechanical impacts. In contrast to ceramic materials, they retain remarkably high transparency, enable flexible fabrication of complex geometries through glass-forming processes such as casting and pressing, and possess lower defect densities and higher homogeneity. These attributes have established glass-ceramics as indispensable materials in demanding technological applications [[4], [5], [6]], including high-precision optics (e.g., astronomical telescope mirrors, laser gyroscope cavities), extreme environment engineering (e.g., high-temperature aerospace

components), precision manufacturing, and consumer electronics. They are thus particularly irreplaceable in harsh environments requiring both high mechanical and optical performance.

In the application of glass-ceramics, high-quality joining between materials inevitably emerges as a critical challenge. Traditional technologies such as adhesive bonding, brazing etc. are ill-suited for high-quality welding of glass–ceramic devices due to inherent issues including chemical element contamination, vulnerability to aging, and significant difficulties in matching thermomechanical properties. While optical contact bonding can achieve welding of glass–ceramic components without introducing foreign elements, it suffers from prominent drawbacks such as low bonding strength, extremely strict requirements for sample surface quality, high processing costs, and low efficiency. Therefore, the exploration of advanced welding processes holds significant application value.

The development of ultrafast laser technology has enabled direct space-selective welding of transparent materials including polymers, glasses and crystals [7], [8], [9], [10], [11], [12]]. Leveraging high peak intensity and short timescale for laser irradiation, focused ultrafast lasers can trigger direct energy deposition at the sample interface in a nonlinear regime, inducing localized high-temperature and high-pressure conditions that facilitate direct interfacial fusion. Recent studies demonstrate this technique well meets the demands for high-strength, high-precision, and contamination-free microwelding and sealing of transparent materials [13,14]. Notably, even for semiconductors opaque in the visible range [15,16], high-quality microwelding can be achieved using long-wavelength ultrafast lasers [17,18]. Existing research also shows that this approach can significantly relax the stringent surface quality requirements, enabling welds with shear strengths exceeding tens of MPa under relatively lenient surface preparation and clamping conditions [19], [20], [21], [22], [23]]. However, the quality of ultrafast laser welding is influenced by multiple factors, including laser spatio-temporal parameters, material properties, and surface conditions. Systematic investigation of these coupled parameters is essential for achieving stable high-quality welding. In this regard, spatial beam shaping

was introduced in ultrafast laser welding to modulate the laser focal field and thermal accumulation effects under the high repetition rate regime [24,25]. It achieved remarkable performance in relaxing the focusing condition, energy deposition homogenization, and cavitation damage suppression. In addition, temporal beam shaping was used to generate modulated burst-mode pulse trains in ultrafast laser welding [[26], [27], [28], [29]], and significantly improved the bonding strength and the ability to control over molten pool volume.

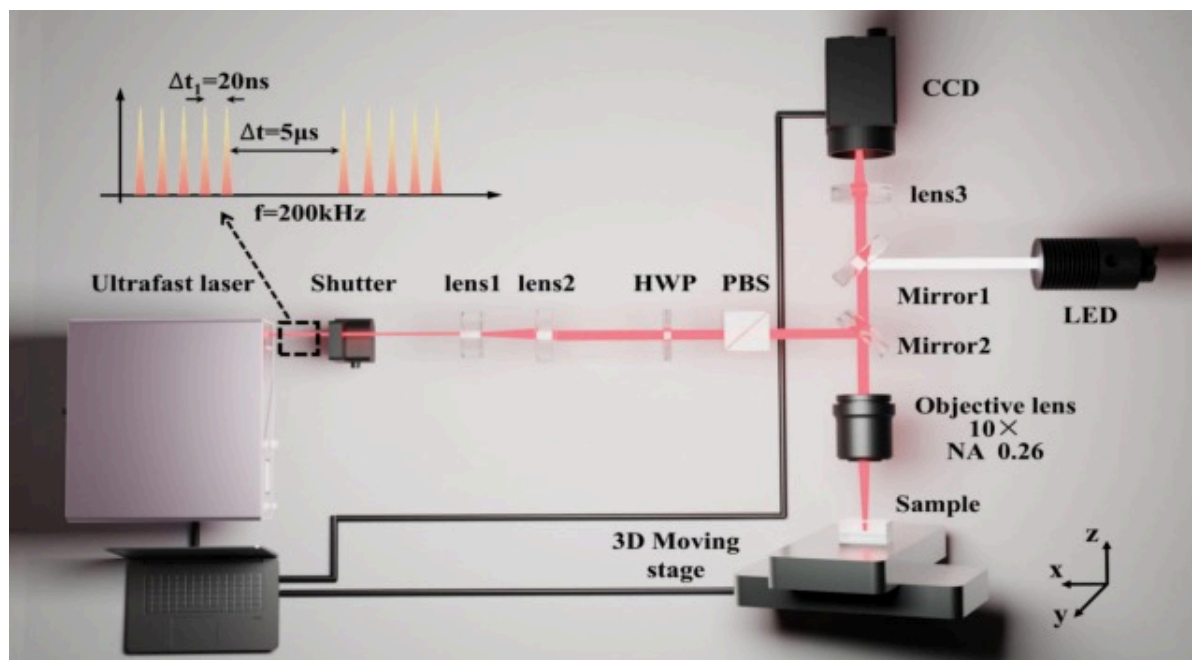
Considering the significant influence of inter-pulse temporal spacing on both the energy deposition and the induced thermomechanical relaxation behavior of materials, this study focuses on the micro-welding of glass-ceramics using a burst-mode ultrafast laser with ns-scale pulse intervals. We discuss the influences of pulse energy, pulse width, and scanning velocity (dictating the linear density of laser pulses) on bonding strength and welding defects. Leveraging the optimizations of laser parameters, we further explore high-strength direct sealing processes for glass–ceramic optical chambers through tailored welding geometry design. This work paves the way for the efficient, high-quality, and cost-effective fabrication and encapsulation of glass–ceramic components in aerospace and precision optical applications.

2. Experiment

The glass–ceramic samples used in the experiments are lithium–aluminum–silicon oxide microcrystalline glass substrates(CDGM, LW1), with a size of $20 \times 10 \times 2 \text{ mm}^3$. The coefficient of thermal expansion in the temperature range 0°C to 50°C is $\sim 1 \times 10^{-8} / ^\circ\text{C}$. The large surface of the samples was mechanically polished to achieve a flatness of half wavelength at 632nm. During the welding process, two samples were stacked and clamped with a fixture to achieve optical contact at the interface. The ultrafast laser (Coherent, Monaco) used in the experiment operated in a burst mode at a central wavelength of 1035nm, with a burst repetition rate of 200kHz, resulting in a time interval of 5 μs between consecutive bursts. Each burst contains 5 laser pulses with a pulse duration adjustable from 1.2ps to 10ps. The time interval between laser pulses within a

burst is approximately 20ns. Considering that both the plasma relaxation and thermal diffusion time scales induced by the ultrafast laser exceed 20ns [30], the laser pulses within a burst produce a more significant cumulative effect during the welding process compared to the intervals between bursts.

To achieve spatially selective fusion of materials at the interface, the ultrafast laser was focused on the sample interface through an objective lens, as shown in Fig. 1, and the movement trajectory and speed of the laser focus on the sample interface were controlled by a three-dimensional translation stage. The pulse energy, defined in the manuscript as the energy of a single pulse within the burst, can be adjusted by means of a half-wave plate and a polarizer. The laser confocal microscope (LCM), phase contrast microscope (PCM), optical transmission microscope (OTM), and scanning electron microscope (SEM) are employed to characterize the material modification to laser excitation on surface and in bulk. To assess the welding quality of the samples, a shear strength testing system described in our previous work [21] was used in the experiments to perform destructive tests. The shear strength is defined by dividing the critical thrust force by the welding-line-enclosed area, which includes both the welding line area and the laser-unaffected regions between welding lines. To evaluate the sealing quality, the laser-sealed glass-ceramic chambers were subjected to leak rate testing.



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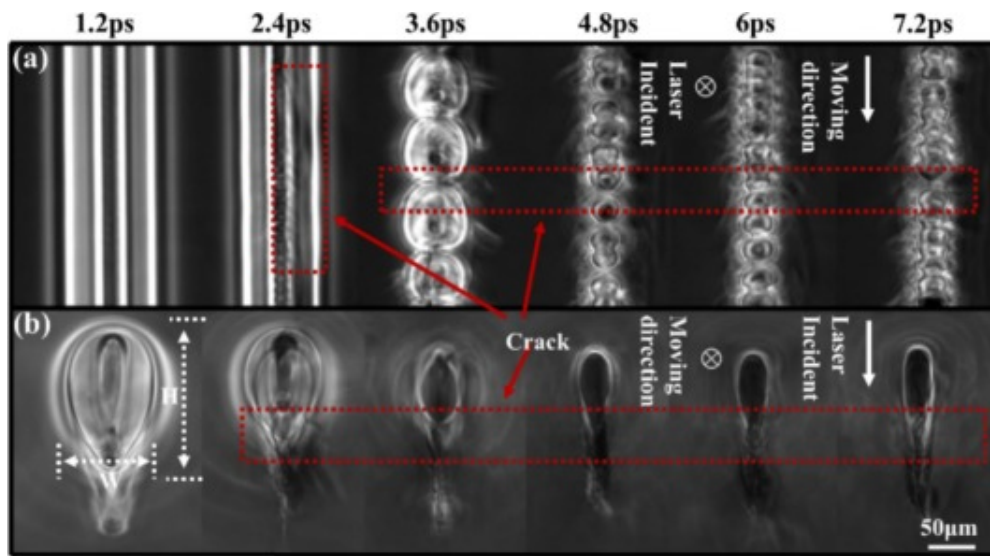
Fig. 1. Schematic diagram of the experimental setup for ultrafast laser burst welding. The 1035nm ultrafast laser with a burst repetition rate of 200kHz is used as the laser source in the experiments. Each burst contains 5 laser pulses with a time interval of 20ns, and the pulse duration is adjustable from 1.2ps to 10ps. The diameter of the laser focus after the 10×objective lens is calculated to be about 4.9μm.

3. Results and discussion

3.1. Impact of pulse duration on material modifications

The influence of inner pulse duration of the burst on laser-induced modifications inside glass-ceramics is first discussed. The burst repetition rate of ultrafast laser was fixed at 200kHz to inscribe embedded lines inside the glass-ceramic sample. The pulse energy

was fixed at $3.0\mu\text{J}$, while the welding speed (i.e., the moving velocity of the laser focus within the sample) was maintained at 0.75mm/s . By controlling dispersion, the laser pulse duration was tuned from 1.2ps to 7.2ps . Using PCM and OTM, we observed the top-views and cross-sections of the laser-modification tracks, as shown in Fig. 2.



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Fig. 2. Modifications inside glass-ceramic samples induced by ultrafast laser bursts with a fixed pulse energy of $3.0\mu\text{J}$ and welding speed of 0.75mm/s . The pulse duration was varied from 1.2ps to 7.2ps . (a) and (b) show the top views (PCM images) and cross-sections (OTM images) of the laser-induced modification lines.

It can be seen that when the inner pulse duration of the burst is 1.2ps , the laser-modified region exhibits smooth and regular melting-and-resolidification characteristics. The cross-section of the modified zone shows a droplet-shaped profile approximately $80\mu\text{m}$ wide and $180\mu\text{m}$ high, significantly exceeding the dimensions of the microscale laser focus. These material modification features align with those produced by conventional high-repetition-rate femtosecond lasers (200fs , $\geq 500\text{kHz}$). This phenomenon primarily

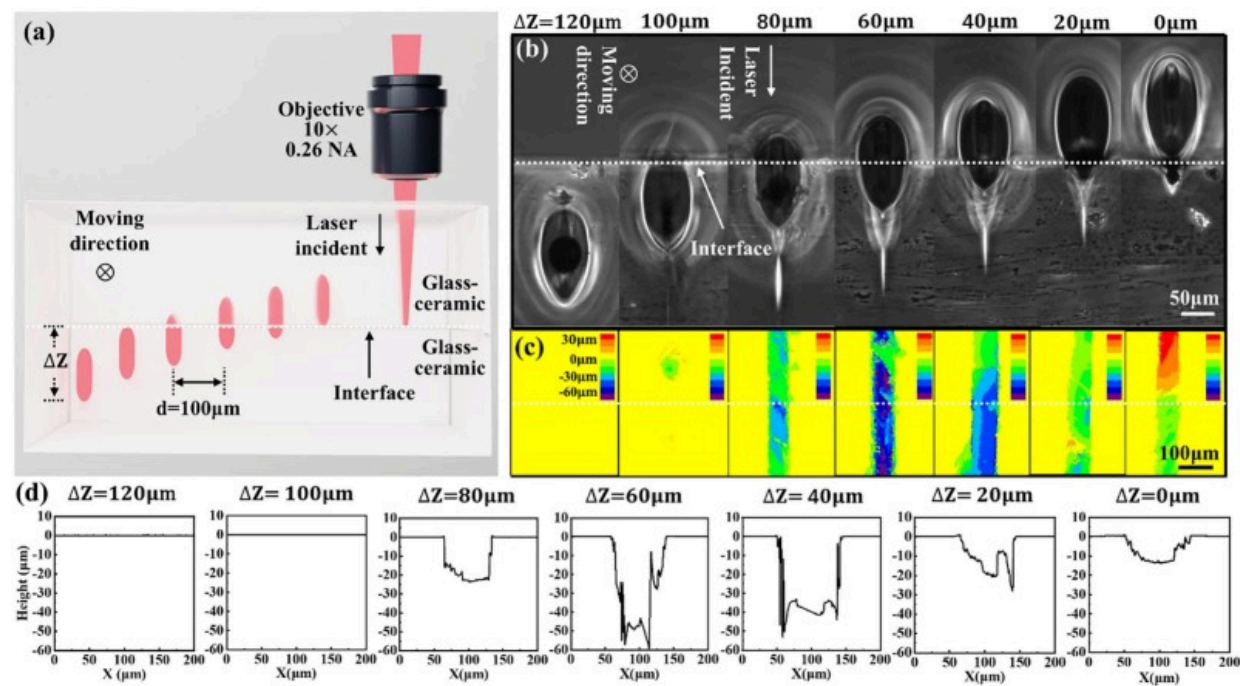
arises from the fact that when a single pulse is modulated into a pulse train spaced by $\sim$ ns intervals, the peak power density of laser irradiation is reduced to some extent. Meanwhile, the cumulative effect is significantly enhanced among the intraburst pulses and influences the laser energy deposition and the temporal evolution of the temperature field [31,32]. Inside the droplet-shaped structure, there is a more elongated region approximately 20 μ m in width, which corresponds to the laser-induced plasma zone or electronic damage [33]. According to the PCM images, the plasma zone exhibits a higher refractive index compared to the surrounding thermal affected zone. Overall, there is no apparent cavitation or crack defect generated in the laser-modification track.

When the pulse duration increases to 2.4ps, the thermal melting characteristics of the material along the laser-modification track become non-uniform, with microcrack defects emerging outside the thermal melting zone. When the pulse duration further increases to above 3.6ps, a large number of bubble defects accompanied by microcracks are generated along the laser-modification track. Meanwhile, the width of the laser-modification track is significantly reduced compared to that at 1.2ps. This indicates that an increase in pulse duration leads to more localized but intense modification of the material. The OTM images show that the formation of bubble and crack defects results in poor light transmittance of the laser-modification tracks induced under longer pulse durations. Therefore, to improve the welding quality, in subsequent experiments, we focus on using the 1.2ps laser burst to perform selective welding and sealing of glass-ceramics.

3.2. Optimization of ultrafast laser welding processes

To determine the optimal focus position for glass–ceramic welding, we first conducted a scan of the laser focus position along the propagation direction. The laser pulse duration was fixed at 1.2ps, with the pulse energy and scanning speed set to 3.0 μ J and 0.75mm/s, respectively. By adjusting the defocus value ΔZ (defined as the distance between the laser focus and the sample interface), as shown in Fig. 3(a), we performed welding lines on the glass–ceramic interface under various defocus conditions. When the welded glass-ceramics undergo shear separation, debris torn from one piece will adhere to the other

(or vice versa). There is a positive correlation between welding strength and the size/thickness of the adherent debris—higher strength corresponds to larger and thicker debris. Thus, by acquiring three-dimensional topographic information of the fractured surfaces after shear separation using a laser confocal microscope, we can correlate the surface undulation height of the weld zone with welding strength, enabling convenient qualitative assessment.



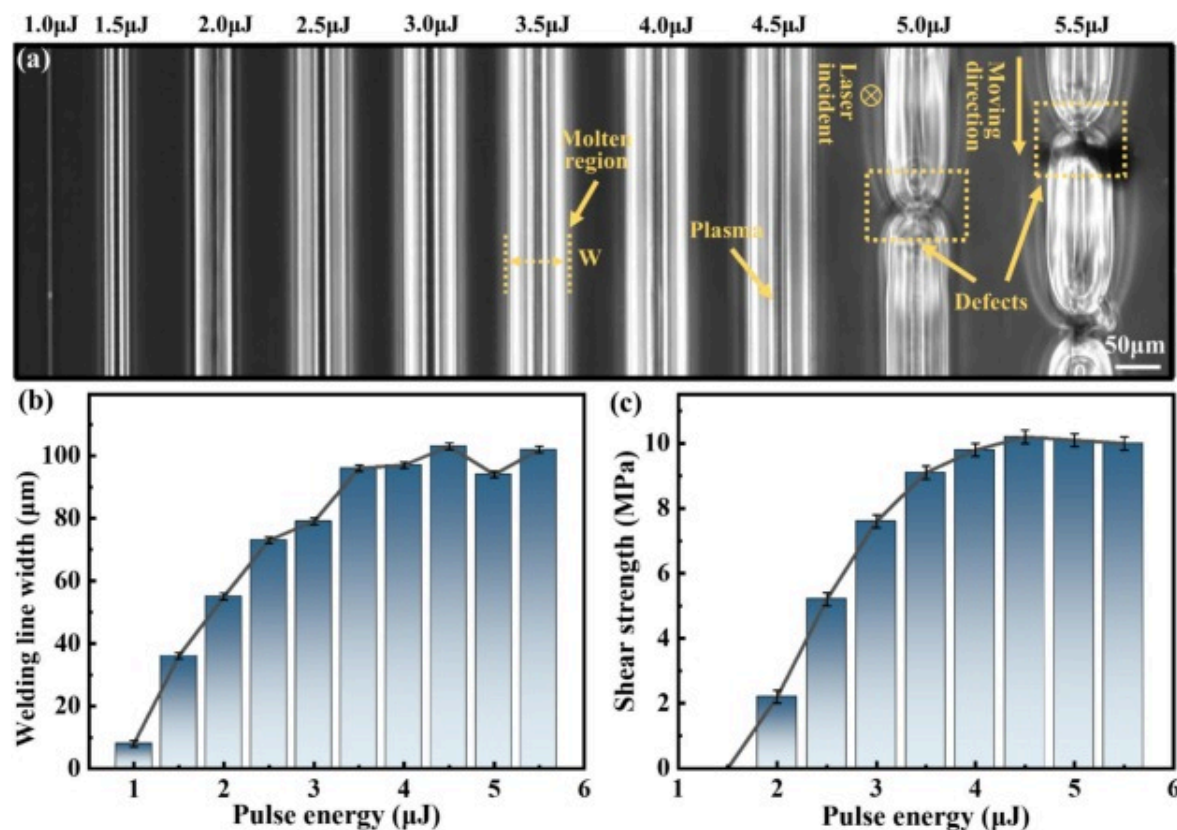
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Fig. 3. (a) Schematic of glass–ceramic sample welding using 1.2ps, 3.0 μJ ultrafast laser burst at a fixed welding speed of 0.75mm/s under varied defocus conditions. (b) cross-sections of the welded lines observed via OTM; (c) 3D topographic maps; and (d) 2D profile curves of the sample interface after shear separation.

Fig. 3(b) shows cross-sectional views of welded samples under different defocus conditions. It is evident that for defocus values ΔZ ranging from $100\mu\text{m}$ to $0\mu\text{m}$, the laser-induced modification region extends through the interface of the welded samples. However, the surface morphology (Fig. 3(c-d)) of samples after shear separation reveals that no effective weld was formed at a defocus value of $100\mu\text{m}$. This is likely due to insufficient fusion across the interface, caused by lower temperatures at the edges of the thermally melted region. Based on measurements of debris thickness, setting the defocus value to $\Delta Z = 60\mu\text{m}$ achieves the highest welding strength.

To identify the optimal laser pulse energy for welding, we performed a scan of pulse energies ranging from $1.0\mu\text{J}$ to $5.5\mu\text{J}$, with the defocus value and welding speed fixed at $60\mu\text{m}$ and 0.75mm/s , respectively. Fig. 4(a) shows top views of the welding lines obtained under different laser pulse energies. It can be observed that at a pulse energy of $1.0\mu\text{J}$, the welding line exhibits no obvious signs of material melting. As the pulse energy increases, the thermal melting features become increasingly distinct, with a significant increase in welding line width, as shown in Fig. 4(b). When the pulse energy exceeds $3.5\mu\text{J}$, the expansion of the thermal melt region saturates, and the welding line width stabilizes at $\sim 100\mu\text{m}$. To evaluate the shear strength of the welded samples, we conducted shear separation tests on samples with a fixed welding area of 16mm^2 and a $150\text{-}\mu\text{m}$ spacing between adjacent welding lines. The test results in Fig. 4(c) reveal that the shear strength of the glass-ceramic samples follows a similar trend with changes in pulse energy, peaking at over 10MPa at $4.5\mu\text{J}$. This indicates that appropriately increasing the width of the thermally melted region helps expand the effective fusion area at the interface, thereby enhancing shear strength. The slight decrease in shear strength observed at pulse energies above $5.0\mu\text{J}$ is attributed to the formation of welding defects caused by excessive thermal effects, as supported by the microscopic images in Fig. 4(a). Consequently, from the perspectives of defect suppression and welding strength enhancement, the optimal laser pulse energy is determined to be $4.5\mu\text{J}$.



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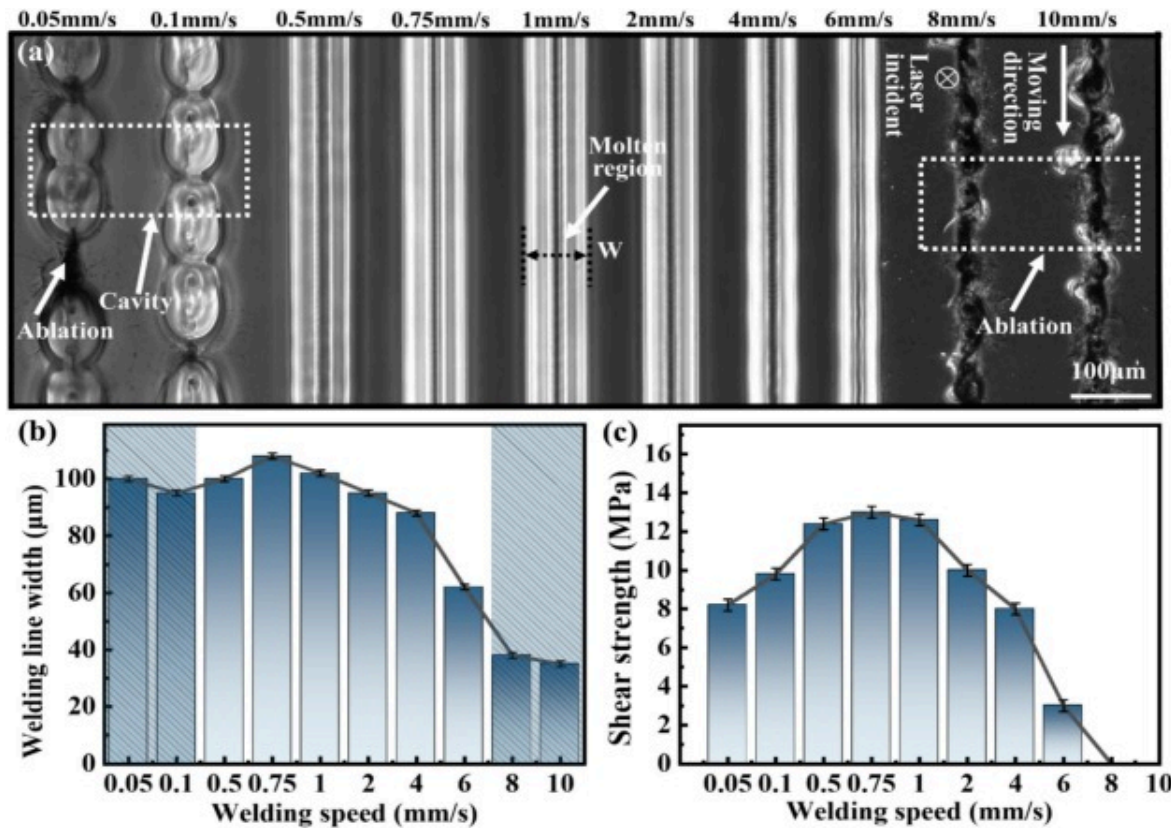
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Fig. 4. Glass-ceramic samples welded using ultrafast laser bursts with a fixed pulse duration of 1.2 ps and welding speed of 0.75 mm/s, where pulse energies varied from 1.0 μJ to 5.5 μJ . (a) Top views (PCM images) of the laser welding lines; (b) Measured width of the laser welding lines under different pulse energies; (c) Shear strength of the laser-welded samples under different pulse energies, with a fixed welding area of 16 mm² and a 150- μm spacing between adjacent welding lines.

Furthermore, we investigated the influence of welding speed on weld quality. In the experiments, the laser pulse energy and defocus distance were fixed at 4.5 μJ and 60 μm , respectively, with a pulse width of 1.2 ps. Since the repetition rate of the burst was

maintained at 200kHz throughout the welding process, changes in welding speed essentially affected the number of laser bursts deposited within unit length.

Fig. 5(a) shows top views of the welding lines obtained with welding speeds ranging from 0.05 mm/s to 10 mm/s. At speeds below 0.1 mm/s, the tracks exhibited quasi-periodic porosity, interfacial ablation, and material expulsion. This is attributed to excessive laser burst deposition per unit length. Given that the repetition rate of the laser bursts was fixed at 200kHz during the welding process, the number of laser bursts deposited per unit length can be up to 4000/ μm at a welding speed of 0.05 mm/s. An apparent thermal accumulation effect between laser bursts is triggered: when the next laser burst reaches the focus, the temperature in the focal region has not yet dropped to room temperature. As the number of bursts increases, the focal region temperature rises stepwise with each incoming burst, eventually exceeding the material's melting point and thereby resulting in supra-threshold heat fluxes that induce shock stress and bubble formation. As the speed increases beyond 0.5 mm/s, these defects were progressively mitigated, and the tracks displayed uniform melting characteristics. Concurrently, the welding line width was found to decrease exponentially with increasing speed (Fig. 5(b)), consistent with reduced energy deposition per unit length. Notably, at speeds larger than 8 mm/s, the material modification transitioned from controlled melting to interface ablation, accompanied by extensive fragmentation. This phenomenon is hypothesized to arise from the reduced thermal accumulation between laser bursts, because the number of laser bursts deposited per unit length decreases to 20/ μm at a welding speed of 10 mm/s. The thermal accumulation effect between laser bursts is insufficient to raise the baseline temperature (the temperature upon laser burst arrival) above the material's melting point—specifically, the material cannot maintain a temperature above its melting point over a relatively longer timescale. Therefore, significant ablation and brittle fracture will be induced on the welding lines during laser burst excitation.



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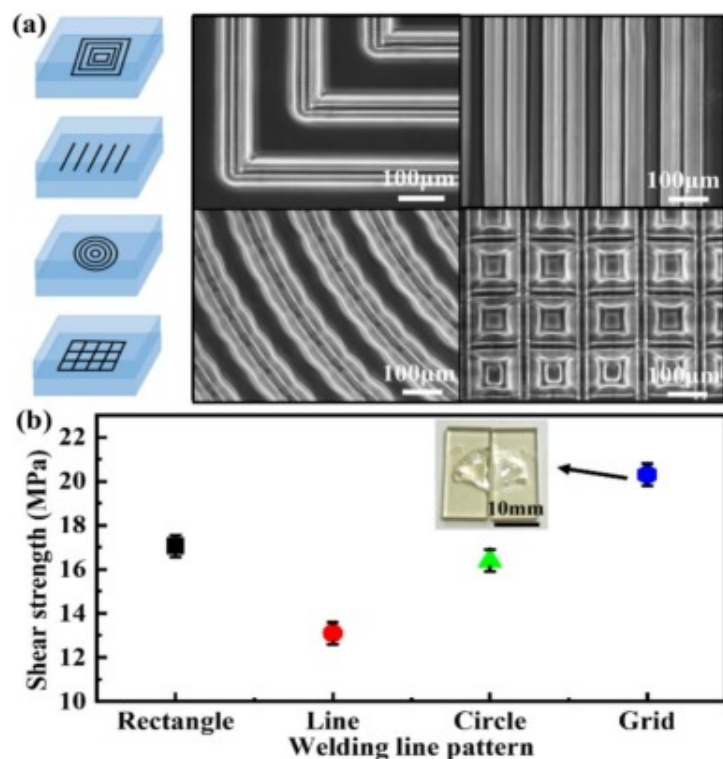
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Fig. 5. Glass-ceramic samples welded by ultrafast laser burst with a fixed pulse duration of 1.2 ps and pulse energy of 4.5 μJ. The welding speed varied from 0.1 mm/s to 10 mm/s. (a) Top views (PCM images) of the laser welding lines; (b) Measured width of laser welding lines under different welding speeds; (c) Shear strength of laser-welded samples under different welding speeds, with a fixed welding region of 16 mm² and a 150-μm separation distance between adjacent welding lines.

The shear strength test results in Fig. 5(c) indicate that welding speed significantly influences the shear strength of glass-ceramic samples. Both bubble defects induced by excessively low speeds and interface ablation caused by overly high speeds lead to

reduced shear strength in the specimens. The welding speed ranging from 0.5 mm/s to 1 mm/s enables the highest shear strength while preserving the optical transparency of the welded zone.

Considering that welding line patterns influence both the welding line density (the welding line area vs. the welding-line-enclosed area) and the residual stress distributions, which could in turn significantly influence shear strength, we employed the burst-mode ultrafast laser with a pulse energy of 4.5 μ J and pulse duration of 1.2 ps to perform glass-ceramic welding with distinct welding line patterns: concentric rectangles, concentric circles, parallel lines, and grid structures. The separation distance between welding lines is fixed at 150 μ m, while the welding speed and defocus distance were fixed at 0.75 mm/s and 60 μ m, respectively. Fig. 6(a) presents schematic diagrams and microscopic images of these different welding lines. Fig. 6(b) presents the measured shear strength of the welded samples. Considering that the area used for calculating shear strength is the welding-line-enclosed area, the value of the shear strength thus reflects the combined action of welding line density and residual stress distribution. Although the grid structure may not be the optimal welding line design for suppressing residual stress, the sample welded with the grid structure was found to have the highest shear strength (exceeding 20 MPa) due to its highest welding line density. Fig. 6(e) shows considerable glass debris torn from the other piece after shear separation, which also reflects strong bonding at the welding interface. This result indicates that when the residual stress is relatively low, the shear strength is more sensitive to welding line density rather than welding line patterns. However, it should be noted that if the spacing between welding lines is further reduced to a small value, the grid structure with higher welding line density will not necessarily result in higher welding strength due to the presence of significant defects and residual stress.



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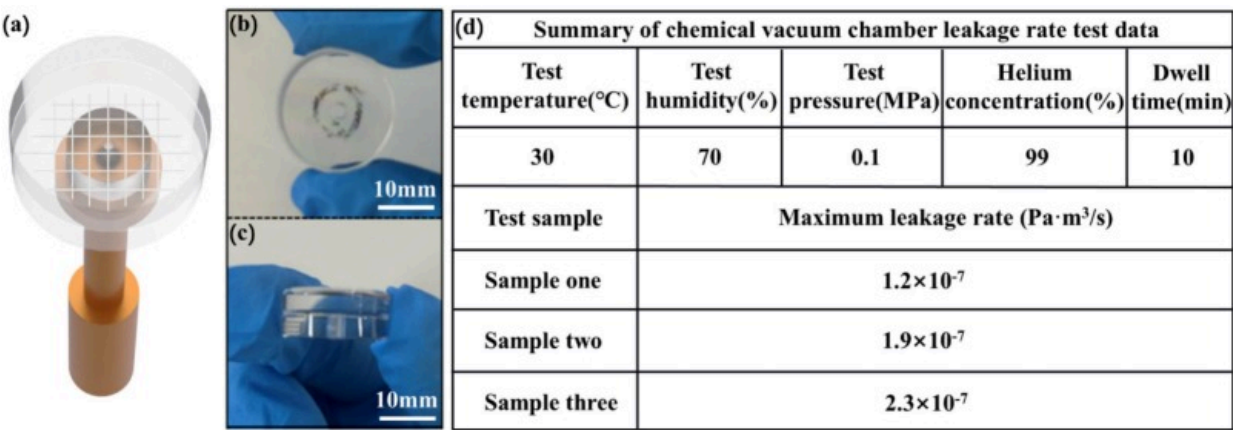
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Fig. 6. Glass-ceramic samples welded by 1.2ps-4.5μJ ultrafast laser burst with distinct welding line patterns: concentric rectangles, concentric circles, parallel lines, and grid structures. The welding speed and defocus distance were fixed at 0.75mm/s and 60μm, respectively. (a) Diagram and PCM images of the welding line patterns; (b) Shear strength of laser-welded samples under different welding line patterns. The inset presents the sample with grid welding lines after shear separation.

3.3. Direct sealing of glass-ceramics devices

As an application, we employed ultrafast laser selective welding to seal one end of a cylindrical glass-ceramic chamber (6mm in diameter) using grid-patterned welding

lines. During the process, the pulse duration and welding speed were fixed at 1.2ps and 0.75mm/s, respectively. The laser pulse energy and defocus value were fine-tuned around the optimized parameters (Sample 1: pulse energy 4.5μJ, defocus value 60μm; Sample 2: pulse energy 4.5μJ, defocus value 80μm; Sample 3: pulse energy 3.5μJ, defocus value 80μm) to ensure optimal sealing quality. Images of the sealed chamber are shown in Fig. 7, Fig. 7. For leak rate testing, a metal gas tube was bonded to the opposite end of the chamber, as illustrated in Fig. 7(a).



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Fig. 7. Glass-ceramic chambers sealed by 1.2ps ultrafast laser burst with grid patterns. Sample 1: pulse energy 4.5μJ, defocus distance 60μm, welding speed 0.75mm/s; Sample 2: pulse energy 4.5μJ, defocus distance 80μm, welding speed 0.75mm/s; Sample 3: pulse energy 3.5μJ, defocus distance 30μm, welding speed 0.75mm/s.

Fig. 7(d) shows the leak rate measurements of the three samples, all of which exhibited excellent hermeticity with values reaching ~10⁻⁷Pa·m³/s. It is worth noting that the surface flatness of the laser-sealed interface was only approximately one wavelength at 632nm, which is significantly lower than the 1/10λ requirement for optical bonding. These results underscore the distinct advantages of ultrafast laser selective welding for

applications requiring high efficiency, strong bonding strength, superior hermeticity, and contamination-free selective sealing of transparent materials.

4. Conclusion

In conclusion, this study represents the first attempt to explore the feasibility of applying ultrafast laser micro-welding technology to the direct selective welding and sealing of glass-ceramics. In the experiments, burst-mode ultrafast lasers were employed, with each burst consisting of 5 ultrafast laser pulses at a time interval of 20ns. Compared to conventional ultrafast lasers, this burst-mode configuration not only effectively mitigates material damage caused by excessively high peak power densities but also moderately prolongs the duration of localized high temperatures—facilitating thorough fusion of interfacial materials.

By scanning the laser pulse duration, pulse energy, welding speed, and welding line structure, we achieved the optimization of glass-ceramic welding. A maximum shear strength of up to 20MPa was obtained under grid-structured welding lines, and this is considered to be the combined action of weld line density and residual stress distribution. Notably, during welding speed scans, excessively high speeds were found to induce significant brittle fracture and ablation in the weld zone, impeding the formation of robust joints. This phenomenon is hypothesized to arise from the following mechanism: an increase in speed reduces the number of bursts accumulated per unit length, the thermal accumulation effect between laser bursts will be insufficient to raise the baseline temperature (the temperature upon laser burst arrival) above the material's melting point—specifically, the material cannot maintain a temperature above its melting point over a relatively longer timescale. Therefore, significant ablation and brittle fracture will be induced on the welding lines during laser burst excitation.

Finally, as a proof-of-principle demonstration, we performed direct selective sealing of cylindrical glass-ceramic optical chambers using grid-patterned weld lines. With the surface flatness of the samples being only one wavelength at 632nm, high-quality sealing

was achieved with a leak rate as low as $10^{-7} \text{Pa}\cdot\text{m}^3/\text{s}$. It is worth noting that previously, it usually required several times the time and cost to polish the sample surface flatness to below one-tenth of the wavelength at 632 nm, plus dozens of hours for high-temperature optical bonding to achieve effective sealing, which lacked spatial selectivity. This work offers valuable insights for practical applications such as laser gyroscope chamber sealing and astronomical optical device welding/sealing in terms of welding efficiency, strength, flexibility, and workpiece requirements.

CRediT authorship contribution statement

Manduo Yu: Writing – review & editing, Writing – original draft, Investigation, Data curation. **Jiachang Deng:** Writing – review & editing, Investigation, Data curation. **Liang Wang:** Writing – review & editing, Writing – original draft, Formal analysis. **Jun Luo:** Investigation, Data curation. **Jiadong Liang:** Visualization, Data curation. **Feng Zhang:** Writing – review & editing, Resources. **Yu Zhang:** Writing – review & editing, Resources. **Wei Zhang:** Writing – review & editing, Resources. **Jiang Wang:** Writing – review & editing, Formal analysis. **Guodong Zhang:** Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis. **Guanghua Cheng:** Writing – review & editing, Supervision, Resources, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that support the findings of this study are available on request.

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